SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Coding Challenge

Course Code:	**DSA0405**	**Course Title:**	**FUNDAMENTALS OF DATA SCIENCE**
CO Assessed:	**CO5-AT2 Coding Challenge**	**Assessment:**	**ASSESSMENT TOOL 2 – Coding Challenge**
Weightage:	**40% of CIA**	**Total Marks:**	**100**
CO4 Statement:	**To understand the concept of supervised and unsupervised methods in machine learning**		
Student Name:	**N HARSHAVARDHAN**	**Reg. No.:**	**192324189**
Batch	**2023**	**Date:**	**27-08-2026**

Code of Conduct

I N HARSHAVARDHAN (192324189) **(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.**

Signature of the Student: N HARSHAVARDHAN

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**	**Marks**
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	

		analyze the case deeply				
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

Code:

```
import numpy as np
from sklearn.linear_model import LinearRegression, Ridge
np.random.seed(10) n = 100
X1 = np.random.randn(n)
X2 = X1 + np.random.randn(n) * 0.1
X3 = np.random.randn(n)
y = 3*X1 + 2*X3 + np.random.randn(n)
X = np.column_stack([X1, X2, X3])
names = ["X1", "X2", "X3"]
print("VIF values:")
for i in range(X.shape[1]):
    other = np.delete(X, i, axis=1)
    r2 = LinearRegression().fit(other, X[:, i]).score(other, X[:, i])
    vif = 1 / (1 - r2)
    print(names[i], round(vif, 2))
def bootstrap(model, X, y, B=200):
    coefs = []
    for _ in range(B):
        idx = np.random.choice(len(y), len(y), replace=True)
        model.fit(X[idx], y[idx])
        coefs.append(model.coef_)
    return np.std(coefs, axis=0)
print("\nOriginal coefficient SD:")
print(bootstrap(LinearRegression(), X, y))
X_reduced = X[:, [0, 2]]
print("\nAfter removing X2:")
print(bootstrap(LinearRegression(), X_reduced, y))
print("\nAfter Ridge regularization:")
print(bootstrap(Ridge(alpha=1), X, y))
```

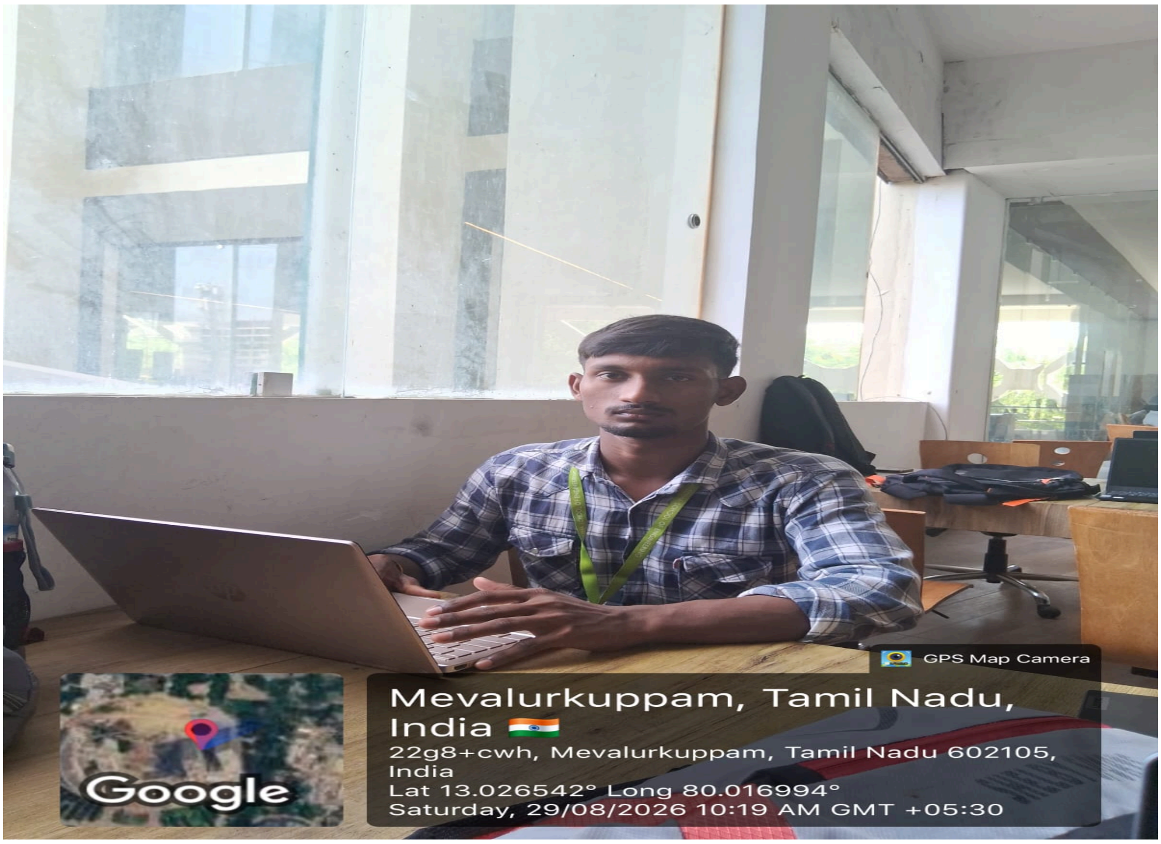
Output:

```
VIF values:
X1 99.99
X2 99.92
X3 1.01

Original coefficient SD:
[1.01649163 1.01302447 0.09869164]

After removing X2:
[0.08350453 0.09163825]

After Ridge regularization:
[0.3135496  0.31891239 0.10192778]
```


GPS Map Camera
Google
Mevalurkuppam, Tamil Nadu, India
22g8+cwh, Mevalurkuppam, Tamil Nadu 602105, India
Lat 13.026542° Long 80.016994°
Saturday, 29/08/2026 10:19 AM GMT +05:30